

Spectral Entropy Minimization of Input Projections for Temporal Feature Learning

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Abstract

Regularization design for temporal feature learning often assumes specific weight geometries facilitate generalization under distribution shift. This study tests whether enforcing a low-entropy singular value spectrum on input projection weights enhances held-out classification accuracy compared to standard weight decay in a subject-disjoint time-series setting. We preregistered a hypothesis that the intervention would exceed a fixed worthwhile margin of 0.0125098 proportion in accuracy gain. Using a Transformer architecture on the UCI Human Activity Recognition dataset, we evaluated the proposed condition against a baseline and two negative controls across 43 primary and 43 confirmatory-replication seeds. The results were inconclusive: the proposed condition improved accuracy by 0.0004419 (95% CI [-0.00213, 0.0031], $p=0.75$) in the primary cohort and 0.002549 (95% CI [-0.000331, 0.0055], $p=0.106$) in the replication cohort. Neither estimate exceeded the preregistered margin, and confidence intervals included zero. These findings suggest that spectral entropy minimization of input projections does not yield robust accuracy gains over standard weight decay in this regime. We report these null results to clarify the mechanistic role of weight spectra in finite-width networks.

1. Introduction

Deep learning models for temporal data frequently encounter distribution shifts when deployed on unseen users or devices, a scenario where standard training objectives often fail to generalize. Regularization techniques aim to mitigate this fragility by constraining the hypothesis space, yet the specific weight geometries that facilitate robust feature learning versus kernel regime behavior in finite-width networks remain poorly understood. While prior work examines entropy maximization for weight noise robustness and singular value decay in Bayesian networks, there is no investigation into whether minimizing the spectral entropy of input projections specifically improves subject-disjoint generalization in time-series transformers. This gap matters for resource-constrained edge AI where subject-disjoint robustness is critical, as clarifying the mechanistic role of weight spectra could guide more effective regularization design. Current regularization strategies often rely on weight decay or sharpness-aware minimization to steer parameters toward flatter loss landscapes [2]. However, these methods operate on the magnitude or local curvature of weights rather than their spectral structure. In sequential tasks, the input projection layer dictates how raw signals are embedded into the model's latent space, making its spectral properties a plausible target for inductive bias. Theoretical work on generalization bounds for autoregressive processes relates empirical loss to denoising loss but does not prescribe specific spectral constraints for input layers [10]. Similarly, while low-rank parameterizations have been proposed to reduce parameter costs when weight matrices exhibit fast singular value decay, the impact of actively minimizing spectral entropy on discriminative accuracy remains untested [14]. This paper presents a preregistered experiment testing whether spectral entropy minimization of input projections enhances feature learning for hard temporal tasks compared to standard weight decay. We address the open problem of what specific weight geometry facilitates feature learning versus kernel regime behavior in finite-width networks. The study employs a subject-disjoint split to ensure evaluation reflects robustness to unseen users, a condition where distribution shifts are prevalent [9]. We report results from both a primary cohort and a confirmatory-replication cohort to ensure reliability. The contributions of this work are:

- We formulate and test a hypothesis that minimizing spectral entropy of input projection weights improves subject-disjoint generalization accuracy beyond standard weight decay. - We provide a rigorous null result from a large-scale seeded experiment (86 total seeds across two cohorts), reporting exact effect sizes and confidence intervals. - We clarify the limits of spectral regularization in temporal transformers, distinguishing between spectral decay properties and active entropy minimization interventions.

2. Related Work

****Out-of-Distribution Generalization in Sequential Data****

Research on generalization in sequential and structured data highlights significant challenges regarding distribution shifts and model capacity. Time series frequently manifest distribution shifts and non-stationary learning dynamics that pose challenges for out-of-distribution generalization [9]. Generalization bounds for autoregressive processes relate empirical loss to denoising loss, characterizing sample complexity in terms of tokens rather than sequences [10]. However, theoretical analysis of generalization in multi-label learning remains in the early stage, with open problems regarding label correlations and inductive biases [7]. When training data is not independent and identically distributed, algorithm performances are challenged by catastrophic forgetting, and regularization-based approaches may fail to discriminate classes from different tasks [17]. These findings collectively establish that standard assumptions often break down in sequential settings, necessitating careful evaluation protocols, though specific claims about unseen users versus random splits require empirical verification beyond survey-level summaries. ****Entropy and Weight Structure in Regularization****

Parallel work examines the role of entropy and weight structure in regularization and robustness. Prior work attempted to test whether explicitly maximizing the entropy of multiplicative weight-noise parameters during training improves held-out classification accuracy on subject-disjoint splits [1]. That study carries a retraction notice because the executed intervention did not implement the preregistered one, so its reported effect cannot be attributed to entropy maximization; it may be cited only as prior work that attempted the question and whose result is not interpretable. Beyond this, the relationship between the entropy of intermediate representations and a model's robustness to distributional shift has been studied, showing that removal of low-entropy bits can benefit out-of-distribution performance [5]. In Bayesian contexts, weight matrices often exhibit fast singular value decay, suggesting low-rank parameterization may be sufficient without standard parameter costs [14]. Regularization characterizes optimal learning in settings where empirical risk minimization fails, with maximum entropy programs exhibiting optimal learners in multiclass settings [6]. Sharpness-Aware Minimization algorithms improve neural network generalization by steering model parameters away from sharp regions, though the underlying mechanisms are not fully understood [2]. Together, this literature suggests entropy and spectral properties are relevant to robustness, but the direction of effect (minimization versus maximization) and the target (weights versus representations) remain distinct questions. ****Projections and Dimensionality Reduction****

Methods involving projections offer alternative perspectives on dimensionality and regularization. Reducing linear program sizes using random and data-driven projections can accelerate solving while maintaining solution quality [3]. In image processing, denoising engines serve as regularization strategies for inverse problems through fixed-point projection [12]. These approaches demonstrate that projection-based constraints can enforce structure, yet their application to spectral entropy minimization in temporal input layers is novel. While text watermarking presents challenges in low-entropy texts, this domain differs significantly from weight spectrum regularization in classifiers [16]. The existing projection literature supports the feasibility of learned constraints but does not settle whether spectral entropy minimization aids temporal feature learning. This work positions itself against these themes by testing a specific spectral intervention on input projections. Unlike prior attempts to maximize weight noise entropy [1], we minimize spectral entropy on deterministic weights. Unlike low-rank Bayesian methods [14], we operate in a standard deterministic training regime with a specific entropy term. We address the gap where no investigation exists into whether minimizing the spectral entropy of input projections specifically improves subject-disjoint generalization in time-series transformers.

3. Method

The experiment tests the hypothesis that enforcing a low-entropy singular value spectrum on input projection weights enhances feature learning for hard temporal tasks compared to standard weight decay. The intervention is predicted to yield higher held-out classification accuracy than the comparator, with mean paired contrast in the hypothesised direction exceeding the fixed worthwhile margin of 0.0125098 proportion. This prediction rests on the assumption that concentrating energy into dominant temporal modes via spectral entropy minimization facilitates more robust feature extraction than distributing energy uniformly, which standard weight decay permits. The core operator is a regularization term added to the training objective that enforces low-entropy singular value spectra on the input projection weights. During each optimization step, the operator computes the singular value decomposition (SVD) of the input projection weight matrix. The resulting singular values are normalized to form a probability distribution. The operator then calculates the Shannon entropy of this distribution and adds it to the loss function with a positive coefficient, such that minimizing the total loss requires minimizing the spectral entropy. This encourages the model to select dominant temporal modes rather than distributing energy uniformly across all singular vectors. The comparator neutralizes this specific spectral pressure by applying standard weight decay without the spectral entropy term, isolating the effect of the spectral constraint from general regularization. The hypothesis targets the mechanistic role of weight spectra in generalization. If spectral entropy minimization improves accuracy, it suggests that finite-width networks benefit from input layers that act as selective filters rather than broad projectors. Conversely, if the intervention yields no gain, it implies that standard weight decay already suffices to manage spectral properties or that spectral entropy is not the critical factor for subject-disjoint robustness in this architecture. The preregistered direction was refuted if the mean contrast fails to exceed the margin or if the confidence interval includes zero. We report both primary and confirmatory-replication cohorts separately to distinguish initial findings from replicated evidence. The method avoids conflating spectral entropy with weight magnitude. Standard weight decay constrains the norm of weights but does not constrain the distribution of singular values; a matrix can have small norm but high spectral entropy (uniform singular values) or low spectral entropy (one dominant singular value). By comparing the proposed condition against a weight decay baseline, we isolate the spectral distribution effect. The negative controls further rule out confounding factors such as optimization instability or arbitrary loss term addition. This design ensures that any observed effect can be attributed to the spectral entropy minimization rather than auxiliary training dynamics.

Operator specification. `SpectralEntropyRegularization` is applied to `input_proj.weight`. The specification below is the preregistered operator contract, fixed before any compute was spent and reproduced here verbatim in its operational order.

- Scope: Singular value distribution of the weight matrix..
- Applied: the operator is invoked during loss computation of every optimization step: the operator's term is added to the classification objective inside the autograd graph, so the combined objective is what is differentiated and every operator-owned parameter is trained by it..
- Ordering: Singular values sorted descending by SVD routine..
- Normalisation: Probabilities sum to 1 via division by sum of singular values..
- Numerical safeguards: Epsilon $1e-8$ added to sum and log argument to prevent division by zero and $\log(0)$..

Algorithm 1 gives the operator in full.

1. Retrieve target parameter tensor W (shape $[64, 9]$).
2. Compute singular values $\sigma = \text{svd}(W, \text{compute_uv}=\text{False})$.
3. Compute $\text{sum_}\sigma = \text{sum}(\sigma) + 1e-8$.
4. Normalize probabilities $p = \sigma / \text{sum_}\sigma$.
5. Compute entropy $H = -\text{sum}(p * \log(p + 1e-8))$.
6. Scale term: in the training objective $= 0.01 * H$.
7. Return in the training objective to harness for addition to objective.

The following invariants hold throughout:

- H is non-negative
- H is maximized when all singular values are equal
- H is minimized when one singular value dominates

Control arm. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs (loss term). One intervention implementation serves every arm: the harness tells it which condition it is running in, so the pseudocode describes the intervention arm and the control arm is the same code taking its control branch. If the condition the harness passes in is a control condition, return here: add no term to the training objective, and leave the registered target parameter and that parameter's optimizer momentum buffer exactly as received. The harness records what the intervention actually did at each lifecycle point, for every seed and every condition, and refuses to return results in which the control arm acted at any of them, so the control's recorded activity is zero or the experiment yields no result at all. Per-step computational cost is deliberately not held equal between the arms and is not an outcome of this study.

4. Experimental Setup

The study design employs a subject-disjoint split to test robustness to unseen users, as standard random splits may hide failure modes specific to distribution shift. Pairing every condition on one seed buys comparability across conditions by controlling for random initialization and data ordering effects, though it costs independence between condition estimates within a seed. We use 43 primary seeds and 43 disjoint confirmatory-replication seeds to ensure sufficient power to detect the preregistered worthwhile margin. The identities of these seeds are recorded in the appendix; we refer to the counts in the prose. This many seeds allows us to construct bootstrap confidence intervals over the paired contrasts, providing a robust estimate of uncertainty beyond single-split evaluations. The first negative control rules out the possibility that any regularization term improves performance simply by acting as a noise source during optimization. The second negative control rules out the possibility that the computational overhead of the SVD operation inadvertently alters training dynamics in a way that benefits accuracy. Without these controls, an improvement in the proposed condition would be indistinguishable from generic regularization effects or implementation artifacts. The analysis computes the mean held-out classification accuracy for each condition and the paired contrast between the proposed condition and the baseline. We report 95% bootstrap confidence intervals over seeds for all metrics. The outcome is defined as the number of correct activity predictions divided by 2,947 windows in the official subject-disjoint test split. These configuration details are fixed across all conditions to ensure that differences in accuracy arise from the regularization term rather than architectural or hyperparameter variations. The design prioritizes internal validity by holding all factors constant except the spectral entropy term. Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at $1e-6$) computed over the TRAINING split only, so no held-out statistic reaches training. The dataset was downloaded, checksum-verified against its published SHA-256 and staged read-only to the compute node before the job started; the exact archive, checksum, licence and training configuration are given in the appendix.

5. Results

We report results for the primary and confirmatory-replication cohorts separately, stating each preregistered decision. In the primary cohort, across 4 conditions and 43 seeds, the proposed condition achieved a mean held-out classification accuracy of 0.9053 (95% CI [0.902, 0.908]), while the baseline achieved 0.9049 (95% CI [0.901, 0.908]). The negative controls performed comparably to the baseline at 0.9049 (95% CI [0.901, 0.908]). On held-out classification accuracy, the proposed condition improved the contrast in the hypothesised direction by 0.0004419 versus the baseline condition (95% CI [-0.00213, 0.0031], paired randomisation $p=0.75$, Cohen's $d_z=0.05$). This difference is not statistically significant as the 95% CI includes zero. The preregistered decision was

inconclusive because the contrast did not exceed the fixed worthwhile margin of 0.0125098 proportion. Figure 1 displays the final metric values per condition with 95% bootstrap confidence intervals over seeds for the primary cohort, illustrating the overlap between the proposed condition and the baseline. Figure 2 shows the mean held-out classification accuracy over training steps for each condition, indicating that learning curves tracked closely throughout training without divergence. In the confirmatory replication cohort, across 4 conditions and 43 seeds, the proposed condition achieved a mean held-out classification accuracy of 0.9087 (95% CI [0.906, 0.912]), while the baseline achieved 0.9061 (95% CI [0.902, 0.91]). The negative controls matched the baseline at 0.9061 (95% CI [0.902, 0.91]). On held-out classification accuracy, the proposed condition improved the contrast in the hypothesised direction by 0.002549 versus the baseline condition (95% CI [-0.000331, 0.0055], paired randomisation $p=0.106$, Cohen's $d_z=0.25$). This difference is also not statistically significant, with the 95% CI including zero. The preregistered decision was inconclusive. Figure 3 displays the final metric values per condition with 95% bootstrap confidence intervals over seeds for the replication cohort, confirming the pattern observed in the primary cohort. Figure 4 shows the mean held-out classification accuracy over training steps for each condition in replication, again showing parallel learning trajectories. The effect size in the replication cohort ($d_z=0.25$) was larger than in the primary cohort ($d_z=0.05$), but the confidence interval still included zero and fell short of the preregistered margin. We do not pool the cohorts or call a primary-only result replicated; both cohorts independently failed to support the hypothesis. The results indicate that spectral entropy minimization did not yield a robust accuracy gain over standard weight decay in this setting. Registered-report decision: the primary cohort was classified as ****inconclusive**** and the independent confirmatory replication was classified as ****inconclusive**** under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

6. Discussion

The quantitative findings indicate that minimizing spectral entropy of input projections does not enhance subject-disjoint generalization accuracy beyond standard weight decay in this transformer architecture. This result contrasts with the hypothesis that concentrating energy into dominant temporal modes would facilitate feature learning. In the context of prior work, this null result clarifies the limits of spectral regularization. While weight matrices often exhibit fast singular value decay [14], actively enforcing low-entropy spectra via optimization did not translate to discriminative gains. This suggests that natural spectral decay observed in trained networks [14] may be a consequence rather than a cause of generalization, or that weight decay alone suffices to induce beneficial spectral properties without explicit entropy terms. The findings also relate to prior attempts to manipulate weight entropy for robustness. Previous work attempted to test whether maximizing entropy of weight noise improves accuracy but carries a retraction notice rendering its result uninterpretable [1]. Our study implements the intervention correctly but finds no benefit to spectral entropy manipulation in the opposite direction (minimization). This suggests that entropy-based regularization of weights, whether maximization or minimization, may not be the critical lever for robustness in this regime. Work on latent variable entropy shows that removal of low-entropy bits can benefit out-of-distribution performance [5], but our results suggest this does not extend to input weight spectra. The distinction between representation entropy [5] and weight spectral entropy appears meaningful; constraining the input projection did not replicate the benefits observed in latent space masking. Regularization characterizes optimal learning where empirical risk minimization fails [6], and sharpness-aware methods steer parameters away from sharp regions [2]. However, our results suggest that spectral entropy is not the relevant geometry for this benefit in temporal transformers. The subject-disjoint split ensures evaluation reflects robustness to unseen users [9], yet the intervention failed to improve performance under this shift. This aligns with observations that regularization-based approaches can be challenged when data is not i.i.d. [17]. The null result across both primary and replication cohorts strengthens the conclusion that spectral entropy minimization is not a reliable method for improving accuracy in this setting. Practitioners should prioritize standard regularization over spectral entropy terms for resource-constrained edge AI where subject-disjoint robustness is critical.

7. Limitations

This study is limited to a single dataset (UCI HAR) and architecture (Transformer encoder), which may restrict generalizability to other temporal domains or model classes. The spectral entropy term was applied only to the input projection weights; applying it to deeper attention layers might yield different results. The worthwhile margin was fixed preregistered, but smaller effects might still be practically relevant in specific high-stakes applications. The experiment used deterministic training with a spectral regularizer; combining this with Bayesian methods or ensembles was not explored. Finally, while subject-disjoint splits test robustness to unseen users, other distribution shifts such as sensor drift or activity definition changes were not evaluated. Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed `uci_har_small_transformer_v1` and to the preregistered smallest worthwhile effect of 0.0125098 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

8. Conclusion

This study tested whether minimizing spectral entropy of input projection weights improves held-out classification accuracy compared to standard weight decay. The results were inconclusive: the proposed condition did not exceed the preregistered worthwhile margin in either the primary or replication cohort, and confidence intervals included zero. This finding changes practice by indicating that spectral entropy minimization is not a necessary addition to standard weight decay for subject-disjoint robustness in temporal transformers. Future work should investigate whether spectral constraints on deeper layers or in combination with other regularization techniques might yield benefits, or whether spectral properties are merely epiphenomenal to generalization in this regime.

Appendix

Executed configuration. The reported run executed the calibrated testbed

``uci_har_small_transformer_v1`` exactly as registered.

- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.

- ``uci_har_smartphones`` —

<https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip> (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdb81031, 61005872 bytes, CC BY 4.0)

- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at $1e-6$) computed over the TRAINING split only, so no held-out statistic reaches training.

- Model: `PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6))`.

- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).

- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.

- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

Seed identities. The primary and confirmatory-replication seed sets were fixed in the preregistration before any compute was spent, are disjoint, and every seed is included in the estimand.

- Primary (43): 2123072646313131395, 2876870697341558572, 1626188215202879716,

1510276966388459924, 731046012090665773, 3260383066045027195, 3095946781025322130, 4381965775545884727, 2805598060171360936, 1017958683207416740, 4042020326231746635, 4023539511489529628, 1871502529958763706, 2254279501808974253, 1144235192351947965, 3712892738658554895, 3406089707180909076, 4440012577721720143, 3980041306974201770, 3346735451993638316, 865312341983968708, 2639610289791772578, 1334636369082240328, 2743053750536161936, 2454452483814521952, 4376507013480164420, 3378328928367937002, 2235759430064723176, 335095552699722741, 3582843840639648566, 1697868845032105404, 2064913684831631250, 558052755835398650, 2501126167359087503, 2998282804321717818, 3159482636166201431, 4144105613280466035, 328700485333559270, 3987679676988384734, 1291110757824769602, 4442649950213859983, 54147398821064359, 1514398810589347120.

- Confirmatory replication (43): 2760814681199202148, 4276509618210360377, 4507730695268138586, 1775243207192301356, 4402558683031212456, 1914937847910204359, 4447932364116798623, 3266695852116340762, 2639495877875133721, 684166292893318705, 763076020713057821, 2284212100604222399, 623189800426898552, 1123976993867976454, 4384758150021590169, 4172343417179275050, 4013643719985881790, 3322921880621971031, 4392335637682857782, 1139371421315206809, 1501434226905974955, 1402960692633542134, 1690201651464576548, 784607708796821782, 4022028640566973239, 1679480667235915961, 1260732336854904037, 1879607340802654956, 3030884773609833226, 658532294107441849, 1198846548640802872, 524796093625990818, 620846012044359089, 3920508414082679309, 731968606963389284, 3744371430145846942, 3914696473985276147, 983119102059608204, 3521400878800560280, 1161765677263725251, 4540894300136491864, 2010139038338001362, 165483925685987617.

Reproducibility

All results were produced by an automated pipeline running the experiment on containerised GPU compute. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). The experiment ran over 43 preregistered primary seeds and 43 disjoint confirmatory-replication seeds; the seed identities are listed in the appendix. Compute job id: 15688229. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbc24249141b15ff622755. The exact code, container reference, seeds, run logs, job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: <https://github.com/ABS-gmbh/ai-research-journal-papers/tree/2c246d256841579152aa0d95a81805b9c14e2653/papers/454ed5f0-7550-46f0-a993-2d3148c93152>.

Ethics

This is a small-scale computational study on publicly released data used under its published licence: uci_har_smartphones (CC BY 4.0). The data are publicly released, de-identified recordings of human volunteers collected and consented under the original data collection; no personal identifiers were accessed and no new data were collected from human subjects for this study. It poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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